Ethan Kuzmik

330-414-7171 | ethan.kuzmik@gmail.com | linkedin.com/in/ethan-kuzmik/ | github.com/EpKuz6121

Education

The Ohio State University — Columbus, OH
B.S. in Data Analytics, Arts & Sciences Honors College — *Aug. 2024 – May 2028*

Technical Skills

Languages: Python, R, SQL, Java, C, C++, x86 Assembly, HTML, LaTeX
Frameworks & Libraries: Pandas, NumPy, scikit-learn, Plotly, Jupyter, Transformers, XGBoost, RandomForest
Developer Tools & Cloud: AWS, Azure, Looker, Tableau, PowerBI, Git, Anaconda, Supabase, Snowflake, Databricks

Experience

Enterprise Technologies Intern — Jun. 2026 – Aug. 2026
PGA of America — *Frisco, TX*

- Built a section-benchmarking web application giving all 41 PGA of America sections a single standardized view of their key metrics, with data pipelines from Looker, AWS, and Workday feeding automated yearly updates; expanded tracking to **23 new metrics** — including first-ever aggregated counts — giving sections self-service visibility into numbers they previously had to request from HQ
- Rebuilt the internal PTO system with a full-year calendar view, an optimizer identifying the highest-value windows to take leave, and a Perplexity Sonar agent for date lookups; added a planning interface that books selected days directly into Workday, **increasing overall PTO scheduling by 30%**
- Developed a live event photo dashboard in TypeScript enabling staff to capture and store photos in an exportable database; **adopted for continued nationwide use** at Drive, Chip and Putt qualifiers
- Built FaceAce, a company-wide Wordle-style game for learning coworkers' names, **driving 200 new users on release and a 50% increase** in PGA's internal HUB 2.0 usage

Data Analytics Project Leader — Aug. 2025 – Present
The Ohio State University, Big Data & Analytics Association — *Columbus, OH*

- Led **30 students** through a complete end-to-end data science project, from problem framing to a **final live presentation**, teaching participants with no prior experience
- Delivered targeted written feedback and guided students on debugging, statistical reasoning, and storytelling
- Ran project sessions covering **Python, R, SQL, Tableau, Pandas, NumPy, Jupyter, and Git**

Projects

Consumer Credit & Loan Behavior Analysis | *Python (Pandas, Plotly), Tableau, PowerPoint* — Apr. 2025

- Analyzed loan behaviors across age groups for 36–60-month terms to surface borrowing patterns and average loan amounts, producing marketing insights to help businesses target demographics and predict lending outcomes
- **Awarded Best Project** in the OSU BDAA Data Science Project Series

PGA Championship Prediction Model | *Python, Pandas, NumPy, scikit-learn, Jupyter* — May 2026

- Built a Monte Carlo simulation that runs the tournament **thousands of times** to generate player win probabilities, modeling players on strokes-gained categories (off-the-tee, approach, around-the-green, putting)
- Backtested simulated probabilities against historical results and Vegas implied odds to validate the model

MLB Shift-Ban Defensive Impact Analysis | *Python, scikit-learn, XGBoost, LightGBM* — Mar. 2026

- Studied how the 2023 MLB infield shift ban affected defensive value at shortstop and second base, using Gaussian Mixture Model clustering to define defensive archetypes and a Difference-in-Differences framework to isolate the effect
- Trained gradient-boosted models (XGBoost, LightGBM) on player-tracking data; found **shallow-range shortstops lost value while shifted second basemen gained**

Data Cleaner AI Agent | *Python, Transformers, TypeScript, Supabase, Lovable* — Dec. 2025

- Built an AI agent to guide early-stage data exploration by flagging high-null columns, detecting outliers, and summarizing key patterns
- Enabled conversational dataset queries to give non-technical users accessible insights and explain abbreviations

NBA Mismatch Efficiency Analysis | *Python (Pandas, Plotly)* — Jun. 2025

- Analyzed NBA player-tracking data to evaluate how positional and size mismatches affect scoring efficiency, measuring points-per-shot to identify optimal shot-clock windows by position and inform lineup construction